PARASITIC ADEQUATE MODULES FORM A SERRE SUBCATEGORY AND THE QUOTIENT IS QUASI-COHERENT MODULES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The kernel of v is the subcategory of parasitic adequate modules. Same chapter, Lemma lemma-parasitic-adequate, invoked at proof lines 2313-2315. Identifies C as a kernel.
- The kernel of an exact functor is a Serre subcategory, the quotient functor exists and is faithful. Homology, Lemmas homology-lemma-kernel-exact-functor and homology-lemma-serre-subcategory-is-kernel, and homology-lemma-quotient-by-kernel-exact-functor, invoked at proof lines 2316-2328. Give Serreness, the induced functor and faithfulness.

Source reference. Original lemma 02MQ: sources/stacks-chow/tags.txt, source lines 3453–3453. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06XK: sources/stacks-chow/tags.txt, source lines 8854–8854. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02MS: sources/stacks-chow/tags.txt, source lines 3455–3455. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06ZM: sources/stacks-adequate/pinned-original-source.tex, source lines 2257–2257. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Lemma 0.1. *Let S be a scheme. The subcategory $\mathcal{C} \subset Adeq(\mathcal{O})$ of parasitic adequate modules is a Serre subcategory. Moreover, the functor v induces an equivalence of categories*

$$Adeq(\mathcal{O})/\mathcal{C} = QCoh(\mathcal{O}_S).$$

Proof. The category $\mathcal{C}$ is the kernel of the exact functor $v : Adeq(\mathcal{O}) \to QCoh(\mathcal{O}_S)$, see Lemma 06ZM. Hence it is a Serre subcategory by Homology, Lemma 02MQ. By Homology, Lemma 02MS we obtain an induced exact functor $\overline{v} : Adeq(\mathcal{O})/\mathcal{C} \to QCoh(\mathcal{O}_S)$. Because u is a right inverse to v we see right away that $\overline{v}$ is essentially surjective. We see that $\overline{v}$ is faithful by Homology, Lemma 06XK. Because u is a right inverse to v we finally conclude that $\overline{v}$ is fully faithful. □